

TASK-2

Data Retrieval & Sorting Dashboard

Aim

To develop a web dashboard that retrieves student records from a database and provides features like sorting (by name/date), filtering (by department), and counting students in each department.

Algorithm

1. Start
2. Connect to the database (student_db).
3. Fetch records from students table.
4. Apply filtering, if department selected and it shows only that department.
5. Apply sorting, sort by name or date of birth.
6. Execute SQL query.
7. Display records in table format.
8. Count students per department using GROUP BY.
9. Display count results.
10. Stop

Program

Dashboard.php:

```
<?php

$conn = new mysqli("localhost", "root", "", "student_db");

$dept = "";

$sort = "";

if(isset($_GET['department'])) {

    $dept = $_GET['department'];
```

```

}

if(isset($_GET['sort'])) {

    $sort = $_GET['sort'];

}

$sql = "SELECT * FROM students WHERE 1";

if($dept != "") {

    $sql .= " AND department='$dept'";

}

if($sort == "name") {

    $sql .= " ORDER BY name ASC";

}

elseif($sort == "dob") {

    $sql .= " ORDER BY dob ASC";

}

$result = $conn->query($sql);

?>

<!DOCTYPE html>

<html>

<head>

<title>Dashboard</title>

<style>

body { font-family: Arial; }

table { border-collapse: collapse; width: 80%; margin: auto; }

```

```
th, td { padding: 10px; border: 1px solid black; text-align: center; }
```

```
.controls { text-align: center; margin: 20px; }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2 align="center">Student Dashboard</h2>
```

```
<div class="controls">
```

```
<form method="get">
```

Filter by Department:

```
<select name="department">
```

```
<option value="">All</option>
```

```
<option>CSE</option>
```

```
<option>ECE</option>
```

```
<option>EEE</option>
```

```
<option>IT</option>
```

```
</select>
```

Sort by:

```
<select name="sort">
```

```
<option value="">None</option>
```

```
<option value="name">Name</option>
```

```
<option value="dob">Date of Birth</option>
```

```
</select>
```

```
<button type="submit">Apply</button>
```

```
</form>
```

```
</div>
```

```
<table>
```

```
<tr>
```

```
<th>ID</th>
```

```
<th>Name</th>
```

```
<th>Email</th>
```

```
<th>DOB</th>
```

```
<th>Department</th>
```

```
<th>Phone</th>
```

```
</tr>
```

```
<?php
```

```
while($row = $result->fetch_assoc()) {
```

```
echo "<tr>
```

```
<td>".$row["id"]."</td>
```

```
<td>".$row["name"]."</td>
```

```
<td>".$row["email"]."</td>
```

```
<td>".$row["dob"]."</td>
```

```
<td>".$row["department"]."</td>
```

```
<td>".$row["phone"]."</td>
```

```
</tr>";
```

```
}
```

```
?>
```

```

</table>

<br><br>

<?php

// Count per department

$count_sql = "SELECT department, COUNT(*) as total FROM students GROUP BY
department";

$count_result = $conn->query($count_sql);

echo "<h3 align='center'>Students per Department</h3>";

echo "<table align='center'>

<tr><th>Department</th><th>Total Students</th></tr>";

while($row = $count_result->fetch_assoc()) {

echo "<tr>

<td>".$row["department"]."</td>

<td>".$row["total"]."</td>

</tr>";

}

echo "</table>";

$conn->close();

?>

</body>

</html>

```

SQL Table:

```
CREATE DATABASE student_db;
```

```
CREATE TABLE students (  
id INT AUTO_INCREMENT PRIMARY KEY,  
name VARCHAR(50),  
email VARCHAR(50),  
dob DATE,  
department VARCHAR(20),  
phone VARCHAR(15)  
);
```

OUTPUT:

Student Dashboard

Filter by Department: Sort by:

ID	Name	Email	DOB	Department	Phone
2	harry	harry37@gmail.com	2026-04-01	ECE	2230065431
1	myth	myth54@gmail.com	2026-04-13	CSE	7766456521

Students per Department

Department	Total Students
CSE	1
ECE	1

Result

Thus, all student records are displayed in a table with features like to sort by name or date of birth, filter by department, and view the count of students in each department.